

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

**SP7-General Insurance Reserving and
Capital Modelling**

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Insurance Company X, is in the business of writing motor and health products. Following an evaluation of the Company's capital adequacy model, the Head of Actuarial team has proposed few improvements. Comment on the appropriateness of the following proposals.

- i) To adopt a fixed percentage charge for operational risk, rather than explicit modelling it. [3]
- ii) To include climate change and emerging environmental risks while modelling capital requirements. [3]

Further, there are certain changes in the Company's strategy. Discuss how would it impact the company's capital requirements by commenting on different risk types.

- iii) To enter into liability line of business with a significant profit commission payable to broker. [3]
- iv) A reduction in reinsurance spending removing lower working layers of individual XOL cover. [3]

[12 marks]

Q.2) Insurer X has traditionally reinsured its liability portfolio using aggregate excess of loss treaty from Reinsurer PQR. The treaty has recently been amended to include a stability clause and a significant increase in aggregate deductible.

- i) Define each of the amended features. [1]
- ii) State the advantages and disadvantages of each feature for both, Insurer X and Reinsurer PQR. [9]

[10 marks]

Q.3) Explain why a general insurance company may need to hold Additional Unexpired Risk Reserve (AURR). [4 marks]

Q.4) A general insurance company, ABCD, is writing property business. Historically, the company has relied on the Basic Chain Ladder method for reserving. However, recent internal discussions suggest that:

- Early development for recent accident years may not be fully credible
- There may be scope to refine estimates using Bornhuetter-Ferguson (BF) techniques
- Discounting reserves could materially impact reported liabilities
- Due to change in the policy's terms and conditions, the underwriter is expecting an IELR of 50% for the latest accident year.

You are a member of the actuarial team and have been asked to perform a review of the reserve estimates and provide insights to the Chief Actuary. The following are the earned premium and cumulative claims paid triangles for the company's property business as on Mar'25.

Accident Year	Earned Premium
2021	15,000
2022	21,000
2023	23,000
2024	26,000
2025	30,000

Accident Year/ Development Year	0	1	2	3	4
2021	4,200	7,100	8,000	8,800	9,000
2022	5,700	9,700	11,000	15,500	
2023	6,900	11,700	12,500		
2024	7,800	13,300			
2025	8,300				

Risk free yield curve:

Year	Risk free rate
0.5	6.40%
1	6.50%
1.5	6.50%
2	6.60%
2.5	6.60%
3	6.60%
3.5	6.65%
4	6.65%

i) Compute the discounted value of reserves, stating any assumptions used. [18]

The Chief Actuary wants to provide a range around the reserve estimate using stochastic methods.

ii) Describe stochastic reserving and state its key features. [4]

iii) Compare the Mack's model and the Over-Dispersed Poisson (ODP) model (no calculation required). [6]

iv) Which model would you prefer for this data and why? [1]

[29 marks]

- Q.5)** Insurance Company XYZ underwrites in property, motor and crop portfolio. Currently, it is in the process of allocating capital.

Line of Business	Standalone SCR (Amount in Rs. crore)
Property	300
Motor	900
Crop	100

Correlation factor between Motor and Crop is 0.25, Motor and Property is 0.25, Property and Crop is 0.50. Total diversified capital for the portfolio = Rs. 1,100 crores.

Explain qualitatively how the allocation would differ using the Shapley method and Proportional method. **[4 marks]**

- Q.6)** Company K is a leading property insurer that has been writing business globally over the last 10 years. Provided below are the results for the last three years: Amount in Rs. crore.

Particulars	2026	2025	2024
Gross written premium	600	500	400
Net written premium	330	230	240
Gross Earned Premium	560	500	400
Net Earned Premium	320	270	250
Gross Incurred Claims	260	450	200
Net Incurred Claims	160	170	120
Operating Expense	40	55	70
Gross Commission	120	80	60
RI Commission	20	25	15

Calculate the key ratios and comment on the company's underwriting performance over the three-year period. **[12 marks]**

- Q.7)** An Insurance Company ABC offers both individual and group financial guarantee products.

- List the various types of financial guarantee products that the Insurer may offer? **[3]**
- State various risks faced by the Insurer for any of the 4 types stated above. **[8]**
- Explain the key differences between individual and group financial guarantee products. **[8]**

[19 marks]

- Q.8)** An Insurance company Z has recently undertaken the following strategy:

- Established a new management team
- Acquired a specialist travel insurer
- Commenced operations in the property insurance business

Discuss with examples any five key operational risks the Insurer is likely to face as a result of these initiatives. **[10 marks]**
